Kadali Tejaswi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.15
Sri Chaitanya	Class XII (CBSE)	2023	96.00%
Sri Chaitanya	Class X (State Board)	2021	100.00%

WORK EXPERIENCE

Research and Development Intern | Murata Machinery, Ltd., JAPAN

- Developed an **automated interview scheduling system** using **Google OR-Tools CP-SAT**, modeling interviewer-student assignments subject to availability, job-role, expertise, and workload constraints.
- Achieved **near-100% assignment rates for 450+ students** with practical solver runtimes of seconds to minutes through **constraint-based optimization and feasibility pre-filtering**.

- Optimized **multi-agent AGV task assignment** using **bipartite matching**, incorporating travel distance, station workload, and **nearby-robot congestion** to reduce warehouse bottlenecks.
- Improved **collision-free multi-agent path planning** using **precomputed BFS distances and PIBT**, addressing deadlocks and queue conflicts and reducing runtime from **minutes to a fraction of a second**.

SKILLS

Languages: C, C++, Python, SQL, JavaScript

Libraries/Frameworks: NumPy, Pandas, SciPy, Matplotlib, Tkinter, React, Vite, Tailwind CSS, Node.js

Developer Tools: Git, GitHub, Docker, VS Code, Jupyter Notebook, Vercel

Web Technologies: HTML, CSS, RESP

Databases: SQL, Firebase, Firestore

Systems: Linux, Windows

Optimization: Google OR-Tools, CP-SAT

PROJECTS

Shuffle -- Club Management Platform

- **Developed** a full-stack club management platform using **React, Vite, Firebase Authentication, and Firestore**, supporting recruitment applications, attendance tracking, member management, and administrative workflows.
- **Deployed** the production application on **Vercel** with Firebase-backed real-time data, authentication, and automated GitHub-based deployments.

Redis-Compatible In-Memory Key-Value Store

- Engineered a **Redis-compatible key-value store** in **C++17** supporting **RESP protocol**, core operations (SET/GET/DEL/EXISTS), **TTL expiration**, **AOF crash recovery**, and **LRU eviction**.
- Designed a **thread-per-connection, 8-shard architecture** with **per-shard synchronization**, handling **100+ concurrent operations** and achieving **6.7K ops/s** (~**25% higher throughput** than the single-threaded baseline).

Sudoku Validation using Multithreading

- Implemented **parallel Sudoku validation using pthreads**, exploring static and dynamic task-scheduling strategies including **Chunk, Mixed, TAS, CAS, and Bounded CAS**.
- Analyzed **thread scheduling and synchronization overhead** by benchmarking execution time across varying Sudoku sizes and thread counts.

Product ID Tracker - Counterfeit Detection & Resolution System

- Developed a **product ID tracking system in C** using a **custom hash table and Binary Search Tree** for efficient insertion, deletion, and lookup.
- Implemented **counterfeit ID detection and resolution** by verifying ID existence and retrieving up to **5 predecessor/successor IDs** from the BST.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications, Operating Systems

Mathematics: Linear Algebra, Probability, Statistics, Calculus, Differential Equations, Complex Variables

POSITIONS OF RESPONSIBILITY

Overall Coordinator- Shuffle (Dance Club, IITH)

- **Leading a 30-member dance club** with 5 Heads and 25 Core members, overseeing performances, workshops, recruitment, and **Inter-IIT participation**.
- **Organized 2 performances and 4 workshops** since the academic year began, while increasing workshop frequency and strengthening the Core team through focused selection.

Coordinator - Diesta (Interdepartmental Sports & Cultural Fest)

- Managed **hospitality, accommodation, refreshments, and transportation** for participants and guests.

SCHOLASTIC ACHIEVEMENTS

Qualified for NTSE Stage 2 (2021) - Among top ~1% students nationwide selected through the State-level screening examination.

EXTRACURRICULAR

Participated in **Group Dance at Inter IIT Cultural Meet 8.0**, representing IIT Hyderabad.

Volunteer, National Service Scheme (NSS).